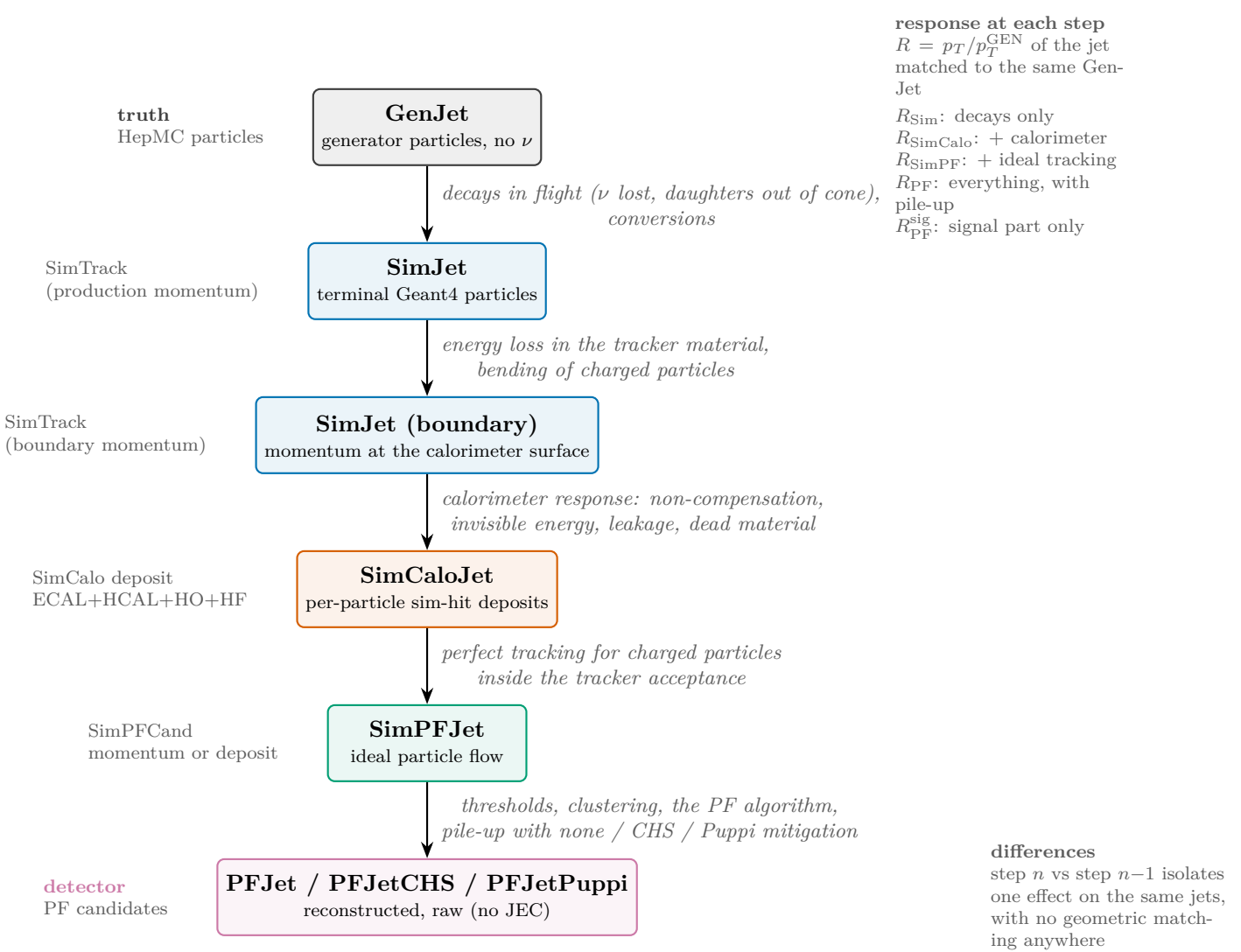

response at each step
$R = p_T/p_T^{\mathrm{GEN}}$ of the jet matched to the same GenJet
R_{Sim}: decays only
R_{SimCalo}: + calorimeter
R_{SimPF}: + ideal tracking
R_{PF}: everything, with pile-up
$R_{\mathrm{PF}}^{\mathrm{sig}}$: signal part only
truth
HepMC particles
GenJet
generator particles, no ν
decays in flight (ν lost, daughters out of cone), conversions
SimTrack
(production momentum)
SimJet
terminal Geant4 particles
energy loss in the tracker material, bending of charged particles
SimTrack
(boundary momentum)
SimJet (boundary)
momentum at the calorimeter surface
calorimeter response: non-compensation, invisible energy, leakage, dead material
SimCalo deposit
ECAL+HCAL+HO+HF
SimCaloJet
per-particle sim-hit deposits
perfect tracking for charged particles inside the tracker acceptance
SimPFCand
momentum or deposit
SimPFJet
ideal particle flow
thresholds, clustering, the PF algorithm, pile-up with none / CHS / Puppi mitigation
detector
PF candidates
PFJet / PFJetCHS / PFJetPuppi
reconstructed, raw (no JEC)
differences
step n vs step n−1 isolates one effect on the same jets, with no geometric matching anywhere